

Karime Martinez | Health and Safety Leader | Data Analytics and Business Intelligence Expert

Colombia | karimemartinezdata@gmail.com | (57) 3332272943 | (57) 3168369555 |

[LinkedIn](#) • [GitHub](#) • [Contáctame](#)



PROFESSIONAL SUMMARY

A visionary Health & Safety Leader and Operations Data Analyst with over 15 years of experience driving operational excellence across large-scale manufacturing, logistics, and construction organizations. Acting as a highly self-managed strategic partner, I optimize processes and automate workflows in agile settings, providing the actionable data and operational insights necessary to support and facilitate C-Suite decision-making. I bridge the gap between robust regulatory compliance (ISO frameworks, high-risk operations), human sustainability, and technological innovation, transforming traditional safety departments into predictive, data-driven strategic assets.

EXPERIENCE

Independent Contractor for Micro 1, Outlier, Alignerr 02/2026 - Present

Subject Matter Expert (SME) in H&S, Data Generalist, AI Trainer, AI Quality Assurance

AI talent and data platforms that connect independent specialists with projects focused on AI model training, evaluation, data quality, and human-in-the-loop validation for the development and improvement of artificial intelligence systems.

- Train, evaluate, and validate AI-generated outputs using subject matter expertise in Health & Safety, analytical reasoning, and structured quality criteria to improve response accuracy, reliability, and consistency.
- Perform human-in-the-loop quality assurance by reviewing AI responses, identifying inaccuracies, inconsistencies, missing context, and reasoning gaps, and providing structured feedback to support model improvement.
- Evaluate AI-generated content against defined regulatory requirements, safety standards, policies, and domain-specific criteria, identifying potential compliance gaps, unsupported statements, and misleading recommendations.
- Apply evidence-based judgment to assess whether AI responses appropriately interpret regulatory and technical requirements, particularly when context, exceptions, or operational conditions may affect the validity of an answer.
- Test and refine AI outputs across different scenarios and prompts, helping establish clearer evaluation parameters and quality standards for AI-assisted applications.
- Support data-related and generalist tasks requiring critical thinking, information analysis, validation, classification, and attention to detail in fast-paced remote environments.
- Work across digital collaboration and productivity ecosystems, including Google Workspace, applying structured information management and analytical workflows to support distributed AI projects.

Colcafé 04/2024 - 03/2026

HSE Coordinator

One of Latin America's leading coffee manufacturers, part of Grupo Nutresa, serving domestic and international markets through large-scale production and supply chain operations.

- Increased budget execution accuracy from 88% to 99% by implementing structured planning, monitoring, and performance tracking mechanisms.
- Standardized operational processes and governance practices across multiple sites, improving execution consistency and accountability.
- Led a team of 9 direct reports and 15 indirect contributors, aligning operational activities with business priorities and strategic objectives.
- Developed KPI monitoring systems and performance dashboards that improved visibility into operational performance and supported data-driven decision-making.
- Improved operational efficiency by optimizing workflows and reducing administrative bottlenecks across key processes.
- Contributed to a 10% reduction in workplace incident rates through performance monitoring, process improvements, and preventive initiatives.

Cementos Argos 07/2022 - 03/2024

HSE Information & Analytics Specialist

Multinational building materials company operating across North America, Central America, the Caribbean, and South America.

- Consolidated operational information supporting decision-making across 19+ countries in the Americas.
- Led the implementation and adoption of a centralized information management platform, improving reporting consistency and data accessibility.
- Designed executive dashboards and KPI reporting systems used by leadership teams to monitor regional performance.
- Standardized reporting methodologies across regions, increasing data reliability and executive visibility.
- Streamlined data collection, validation, and reporting processes, reducing reporting inefficiencies and improving information quality.
- Supported organizational change management initiatives that accelerated technology adoption across multicultural teams.
- Managed regional reporting structures and operational performance data.
- Developed standardized reporting mechanisms and KPI tracking processes.
- Improved information quality and reporting consistency across multiple business units.

Colmena Seguros

Prevention Consultant

12/2020 - 06/2022

- Risk management and occupational health organization supporting companies across multiple industries.*
- Advised organizations on management systems, process standardization, and operational compliance initiatives.
 - Developed performance indicators and monitoring frameworks to support continuous improvement programs.
 - Supported organizations during ISO 45001 implementation and management system integration projects.
 - Designed and facilitated training programs, strengthening internal capabilities and process adoption.
 - Guided organizations in the development of strategic roadmaps for risk management and operational performance improvement.

ARL SURA

Risk Management Consultant

07/2017 - 07/2018

- Leading risk management and prevention organization supporting operational performance across multiple industries.*
- Managed behavior-based improvement programs focused on organizational culture and operational performance.
 - Analyzed behavioral and operational data to identify trends, risks, and improvement opportunities.
 - Delivered insights and recommendations used by leadership teams to strengthen preventive strategies.
 - Designed interventions that improved employee engagement and proactive participation.
 - Supported decision-making through reporting, trend analysis, and performance monitoring activities.

Cementos Argos

HSE Analyst

07/2013 - 06/2017

- Multinational building materials company operating across North America, Central America, the Caribbean, and South America.*
- Supported implementation of enterprise-wide management system transformation initiatives.
 - Coordinated contractor management processes and operational compliance activities.
 - Participated in large-scale plant shutdown planning and execution efforts.
 - Assisted leaders in monitoring operational performance and implementing preventive actions.
 - Supported the deployment of risk management methodologies and operational excellence practices.

Constructora ACH SAS

HSE Supervisor

01/2012 - 05/2013

- Construction contractor delivering project-based services and workforce support for industrial and commercial developments, including operations associated with Zona Franca Parque Central.*
- Project-Based Experience. Early-career experience across construction and industrial project environments, building a foundation in management systems, process implementation, performance monitoring, risk analysis, and operational coordination.
 - Managed operational and HSE activities across 5 simultaneous contracts in a dynamic construction environment.
 - Developed control mechanisms to manage compliance requirements and a high-rotation workforce of 200+ employees.
 - Performed risk assessments and monitored field operations involving high-risk activities, including excavation, work at height, hot work, and hazardous energy.
 - Tracked performance indicators and implemented action plans based on operational results and identified improvement opportunities.

Herrera y Duran Ltda.

Integrated HSEQ Management Systems Leader

06/2009 - 01/2012

- Construction contractor providing industrial and infrastructure services, with experience supporting complex projects and contracts for organizations including Cementos Argos, Ecopetrol/Reficar, and CI Océanos.*
- Project-Based Experience. Early-career experience across construction and industrial project environments, building a foundation in management systems, process implementation, performance monitoring, risk analysis, and operational coordination.
 - Coordinated the implementation of integrated management systems covering occupational health and safety, environmental management, and quality.
 - Developed strategic plans, risk assessments, legal compliance matrices, management objectives, and performance indicators.
 - Supported internal audits, management reviews, corrective actions, and continuous improvement processes.
 - Contributed to the successful implementation and certification of the Occupational Health and Safety Management System under ISO 18001.

EDUCATION & CERTIFICATION

Corporación Universitaria Americana

Master of Business Administration and Management

01/2026 - 06/2027

Corporación Unificada Nacional de Educación Superior CUN

Postgraduate Degree Specialization in Data Analytics, Data Modeling / Warehousing and Database administration

09/2024 - 02/2026

Fundación de Educación Superior San José

Bachelor of Engineering, Industrial Engineer

01/2023 - 12/2024

Corporación Universitaria Latinoamericana CUL <i>Postgraduate Degree Specialization in H&S Management information Systems</i>	06/2021 - 12/2022
Corporación Universitaria Latinoamericana CUL <i>Bachelor of Business Administration, Risk Management Systems Administration</i>	01/2020 - 12/2021

SKILLS & TOOLS

- Strategic & HSE Capabilities: Integrated Risk Governance, ISO 45001/14001/9001/39001, High-Risk Operations Control Coordination and Planning, Operational Governance, Change Management, Process Mapping, Performance Management, SGSST, Compliance Evaluation, Regulatory Analysis, Annual Planning.
- HSE Regulatory & Technical Frameworks: Colombian SG-SST Framework, Decreto 1072 de 2015, Ley 1562 de 2012, Resolución 0312 de 2019, Colombian Occupational Risk System, Legal Requirements Identification & Evaluation, Regulatory Compliance Audits, HSE Legal Compliance Assessment, ISO 45001, ISO 14001, ISO 9001, ISO 39001, ILO-OSH 2001, OSHA Principles, NFPA Standards Awareness, ANSI/ASSP Z10 Awareness, Contractor Safety Management, Permit-to-Work Systems, Risk Assessment & Hazard Identification, Risk Control & Mitigation, Lockout/Tagout (LOTO) & Hazardous Energy Control, Work at Height Safety, Confined Space Entry, Hot Work Safety, Lifting Operations & Rigging Safety, Electrical Safety, Chemical Safety Management, Machine Safety, Industrial Hygiene Fundamentals, Incident & Near-Miss Investigation, Root Cause Analysis, Emergency Preparedness & Response, Fire Prevention & Protection, Road Safety Management. Industrial Maintenance Shutdown Management, Major & Minor Plant Turnarounds, Maintenance Project HSE Planning, Shutdown HSE Coordination, Simultaneous Operations (SIMOPS), Contractor & Subcontractor Coordination, Shutdown Readiness Planning, Pre-Startup Safety Review (PSSR), Project HSE Planning, Construction & Maintenance Project Safety, Operational Readiness.
- Data & Analytics Skills: Business Operations, Operational Intelligence, Data Analytics, KPI Management, Executive Reporting, Workflow Automation, Data Integration, ETL/ELT, AI Training, AI Quality Assurance, AI Evaluation, Data Validation, Data Quality, Human-in-the-Loop, Subject Matter Expertise, Prompt Evaluation, Analytical Reasoning, Critical Thinking, Information Analysis.
- People, Training & Safety Culture: HSE Training & Facilitation, Training Needs Assessment, Training Program Design, Learning Content Development, Training Curriculum Development, HSE Training Plan Development, Annual Training Planning, Competency Management, Competency Gap Analysis, Learning Objectives Development, Instructional Content Design, Technical Training Material Development, Training Modules & Course Development, Knowledge Transfer, Safety Culture Development, Behavioral Safety Programs, Behavior-Based Safety (BBS), Worker Engagement & Participation, Safety Leadership, Leadership Engagement, Behavioral Change, Toolbox Talks & Pre-Job Safety Briefings, On the Job Trainings (OJT).
- Tools: SQL, Python, BigQuery, Excel, Power BI, Tableau, Looker Studio, Google Apps Script, AppSheet, Power Automate, Google Workspace, AWS (Low-Code/No-Code).